Signorini's Problem

Max Stuthmann

March 9, 2026

Content

Introduction

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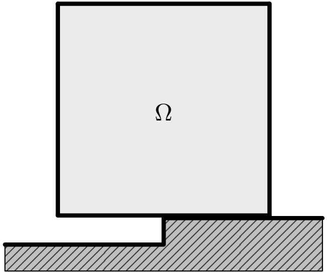

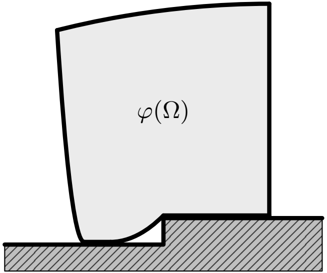

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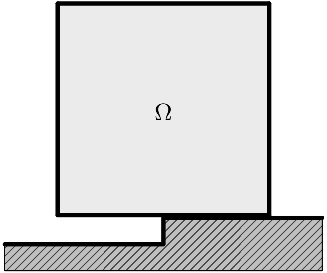

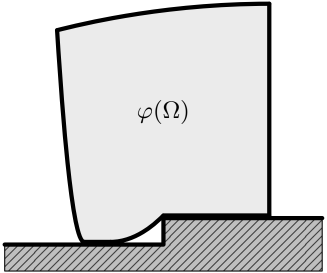

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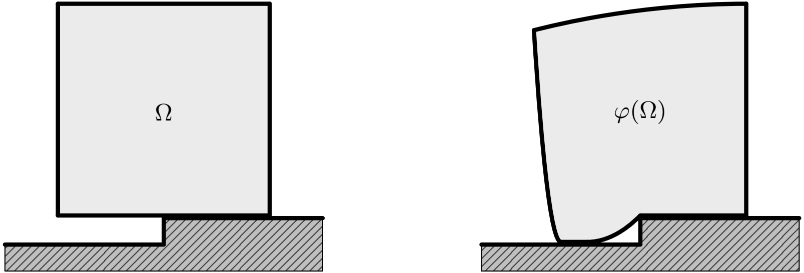

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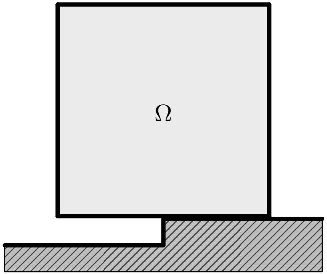

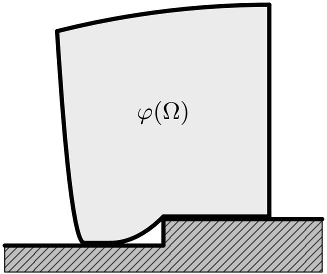

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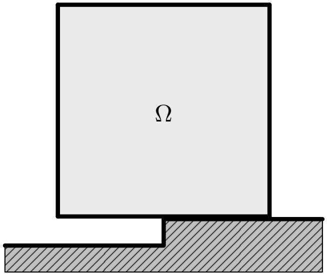

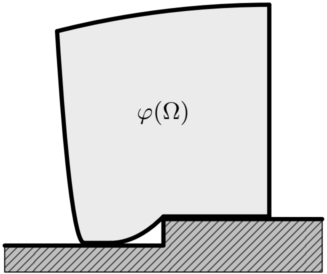

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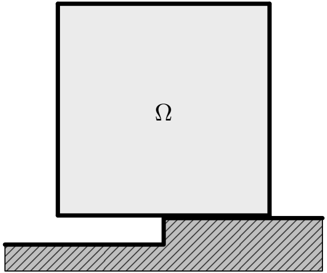
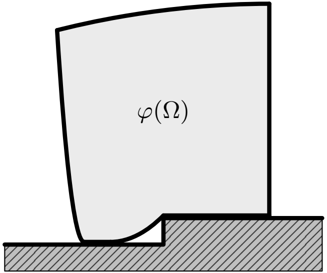
- Rigid foundation can be described by gap-function g but we assume a flat foundation, i.e. $g = 0$.





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- For $k = 1$ we get

$$\|u\|_{1,\Omega} = \left(\|u\|_{0,\Omega}^2 + \|\nabla u\|_{0,\Omega}^2\right)^{1/2}, \qquad u \in H^1(\Omega;\mathbb{R}^m).$$

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- Boundary traces of H^k-functions live in H^s spaces.

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$$H^{s-1/2}(\Gamma; \mathbb{R}^m) := \gamma_0(\mathcal{C}^\infty(\overline{\Omega}; \mathbb{R}^m))$$

and equip it with the norm

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- Same set for all $s > 1/2$ but different norms.

The Trace Theorem

Theorem (Trace Theorem)

Let $1/2 < s \leq 1$. Then the classical trace operator γ_0 extends uniquely to a continuous surjective linear operator

$$\gamma : H^s(\Omega; \mathbb{R}^m) \to H^{s-1/2}(\Gamma; \mathbb{R}^m).$$

In particular, there exists a constant $C_{\mathrm{tr},s} > 0$ such that

$$\|\gamma u\|_{s-\frac{1}{2},\Gamma} \leq C_{\mathrm{tr},s} \|u\|_{s,\Omega} \qquad \forall\, u \in H^s(\Omega; \mathbb{R}^m).$$

In the special case $s = 1$, one has

$$\ker(\gamma) = \overline{\mathcal{D}(\Omega; \mathbb{R}^m)}^{\|\cdot\|_{1,\Omega}} =: H^1_0(\Omega; \mathbb{R}^m).$$

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- Moreover, we define

$$H^{s-1/2}_{00}(\Gamma_0;\mathbb{R}^m) := \left\{ g \in H^{s-1/2}(\Gamma_0;\mathbb{R}^m) \;\middle|\; \mathrm{Ext}_0(g) \in H^{s-1/2}(\Gamma;\mathbb{R}^m) \right\},$$

where $\mathrm{Ext}_0(g)$ denotes the zero extension of g from Γ_0 to Γ with the norm

$$\|g\|_{s-\frac{1}{2},00,\Gamma_0} := \|\mathrm{Ext}_0(g)\|_{s-\frac{1}{2},\Gamma}.$$

Dual Spaces of Boundary Spaces

- We define

$$H^{-1/2}(\Gamma;\mathbb{R}^m) := \big(H^{1/2}(\Gamma;\mathbb{R}^m)\big)', \qquad H^{-1/2}(\Gamma_0;\mathbb{R}^m) := \big(H^{1/2}_{00}(\Gamma_0;\mathbb{R}^m)\big)'.$$

with duality pairings $\langle \cdot, \cdot \rangle_\Gamma$ and $\langle \cdot, \cdot \rangle_{\Gamma_0}$, respectively.

Lifting Operators

Theorem (Lifting Operator)

Let $\Omega \subset \mathbb{R}^d$ be a bounded Lipschitz domain and let $\Gamma_0 \subset \Gamma$ be relatively open.

i) *There exists a bounded linear operator $\mathcal{L} : H^{1/2}(\Gamma; \mathbb{R}^d) \to H^1(\Omega; \mathbb{R}^d)$ such that*

$$\gamma(\mathcal{L}g) = g \qquad \forall\, g \in H^{1/2}(\Gamma; \mathbb{R}^d).$$

ii) *There exists a bounded linear operator $\mathcal{L} : H^{1/2}_{00}(\Gamma_0; \mathbb{R}^d) \to H^1(\Omega; \mathbb{R}^d)$ such that for every $g \in H^{1/2}_{00}(\Gamma_0; \mathbb{R}^d)$,*

$$\gamma(\mathcal{L}g)|_{\Gamma_0} = g \qquad \text{and} \qquad \gamma(\mathcal{L}g) = 0 \text{ on } \Gamma \setminus \Gamma_0.$$

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$H(\mathrm{div};\Omega)$ for Matrix-Valued Fields

- The space $\mathrm{Sym}^2(\mathbb{R}^d)$ of symmetric second order tensors is
$$\mathrm{Sym}^2(\mathbb{R}^d) := \left\{ s \in \mathbb{R}^d \otimes \mathbb{R}^d \;\middle|\; s^\top = s \right\}.$$

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- If $\mathcal{A} : \Omega \to \mathcal{L}(\mathrm{Sym}^2(\mathbb{R}^d); \mathrm{Sym}^2(\mathbb{R}^d))$ is a fourth order tensor and $s \in \mathrm{Sym}^2(\mathbb{R}^d)$, we use the *double contraction*
$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^{d} \sum_{l=1}^{d} \mathcal{A}_{ijkl}(x) s_{kl}, \qquad s : \mathcal{A}(x) : s := \sum_{i=1}^{d} \sum_{j=1}^{d} \sum_{k=1}^{d} \sum_{l=1}^{d} \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- We define
$$H(\mathrm{div};\Omega) := \left\{ \tau \in L^2(\Omega; \mathrm{Sym}^2(\mathbb{R}^d)) \;\middle|\; \mathrm{div}\,\tau \in L^2(\Omega;\mathbb{R}^d) \right\},$$
where $\mathrm{div}\,\tau$ is understood row-wise in the weak sense.

Normal Traces and Boundary Tractions

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- If we assumed $\tau \in H^1(\Omega; \mathbb{R}^{d\times d})$, then $\gamma\tau \in L^2(\Gamma; \mathbb{R}^{d\times d})$ and $(\gamma\tau)\mathrm{n} \in L^2(\Gamma; \mathbb{R}^d)$.

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- On $H(\operatorname{div}; \Omega)$, the trace theorem does not hold. We need a different concept here.

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- In the smooth case, $\gamma_{\mathrm{n}}(\tau)$ coincides with the classical traction $\tau\,\mathrm{n}$. We thus write $\tau\,\mathrm{n} = \gamma_{\mathrm{n}}(\tau)$.

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- The normal trace on Γ_0 is defined as the unique element $\gamma_{\mathrm{n},0}(\tau) \in H^{-1/2}(\Gamma_0; \mathbb{R}^d)$ given by

$$\langle \gamma_{\mathrm{n},0}(\tau),\, g\rangle_{\Gamma_0} := \langle \gamma_{\mathrm{n}}(\tau),\, \mathrm{Ext}_0(g)\rangle_{\Gamma}, \qquad g \in H^{1/2}_{00}(\Gamma_0; \mathbb{R}^d),$$

where $\mathrm{Ext}_0(g)$ denotes the zero extension of g from Γ_0 to Γ.

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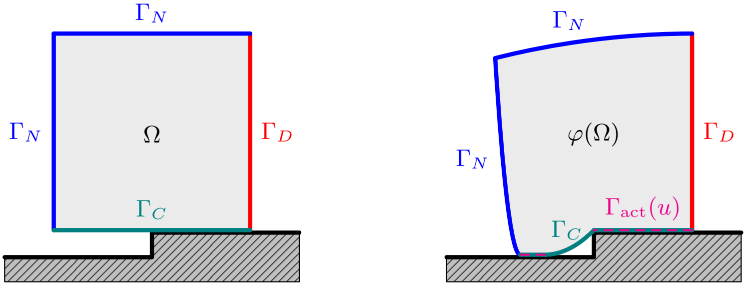
- The body in equilibrium Ω configuration is $\varphi(\Omega)$ with $\varphi(x) = x + u(x)$.

Boundary Decomposition

- Each part has different boundary conditions:



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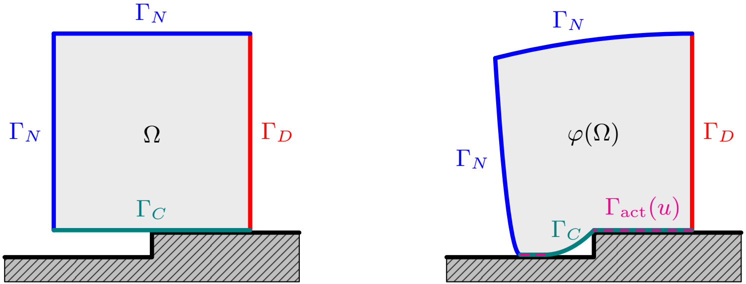

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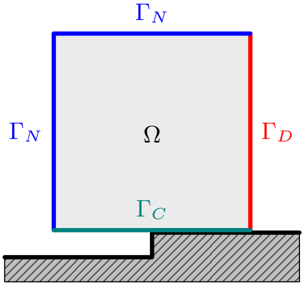

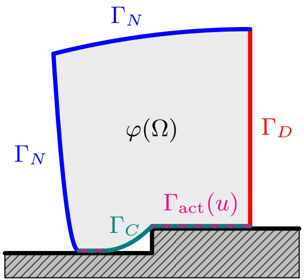

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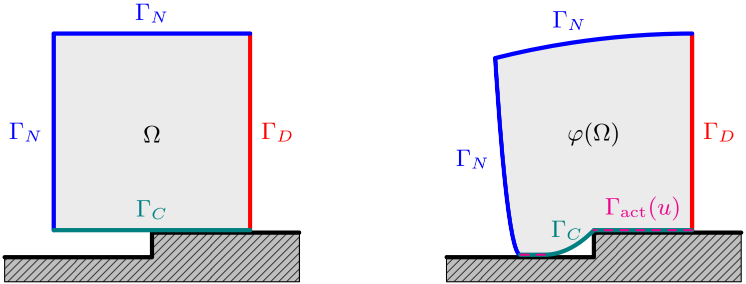

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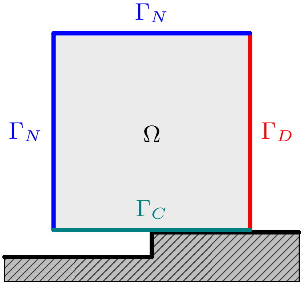

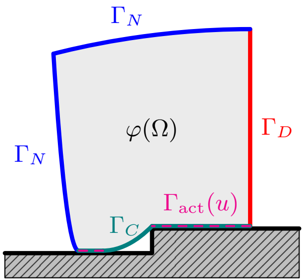

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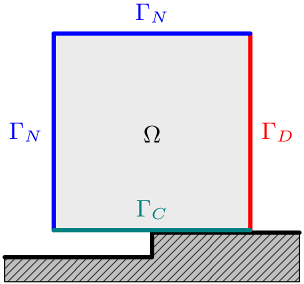

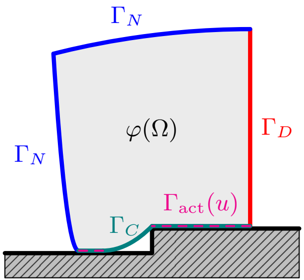

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- K is a cone, i.e. if $v \in K$ then $\lambda v \in K$ for all $\lambda \in [0,\infty)$.

Small Strain Elasticity

- *Small strain tensor*:
$$\varepsilon : H^1(\Omega;\mathbb{R}^d) \to L^2(\Omega;\mathrm{Sym}^2(\mathbb{R}^d)), \qquad u \mapsto \varepsilon(u) := \frac{1}{2}\left(\nabla u + \nabla u^\top\right).$$

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Virtual Work: External and Internal Forces

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- Work $=$ force $\cdot$ displacement. Thus

$$\ell(v) := \int_\Omega f \cdot v \, \mathrm{d}x + \int_{\Gamma_N} F \cdot (\gamma v) \, \mathrm{d}s, \qquad v \in V$$

describes the *virtual work of external forces*.

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Virtual Work: External and Internal Forces

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$$a(u, v) := \int_\Omega \sigma(u) : \varepsilon(v) \,\mathrm{d}x, \qquad u, v \in V.$$

Standard Properties of a and ℓ

Proposition

The symmetric bilinear form $a : V \times V \to \mathbb{R}$ defined above is

i) bounded, *i.e. for all $v, w \in V$ there exists $C_1 < \infty$ such that*

$$|a(u,v)| \leq C_1 \, \|u\|_{1,\Omega} \, \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_1 = C_{\mathcal{A}},$$

ii) V-elliptic, *i.e. for all $v \in V \setminus \{0\}$ there exist $C_2 > 0$ such that*

$$a(v,v) \geq C_2 \, \|v\|_{1,\Omega}^2 \qquad \textit{with} \qquad C_2 = \frac{\alpha_{\mathcal{A}}}{C_K^2}.$$

Moreover, the linear functional ℓ defined above is bounded, *i.e. for all $v \in V$ there exists $C_3 > 0$ such that*

$$|\ell(v)| \leq C_3 \, \|v\|_{1,\Omega} \qquad \textit{with} \qquad C_3 = \max\{1, C_{\mathrm{tr}}\} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N} \right)$$

Energy Minimization

- The *potential energy* of the elastic body is given by the quadratic functional

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Problem (Signorini's Problem – Energy Minimization)

Find $u \in K$ such that

$$\mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v) = \min_{v \in K} \frac{1}{2}\, a(v,v) - \ell(v).$$

Existence of Minimizer

Lemma (Weak lower semicontinuity of the norm)

Let $(X, \|\cdot\|_X)$ be a normed space. Then $\|\cdot\|_X : X \to \mathbb{R}_{\geq 0}$ is a weakly lower semicontinuous functional, i.e. if $x_j \rightharpoonup x$ in X, then

$$\|x\|_X \leq \liminf_{j\to\infty} \|x_j\|_X .$$

Proof: Omitted here.

Lemma ($\mathcal{J}$ is coercive and w.l.s.c.)

The functional $\mathcal{J}$ is

i) coercive *on V, i.e.*

$$\mathcal{J}(v) \to \infty \qquad \textit{as} \qquad \|v\|_{1,\Omega} \to \infty,$$

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- By boundedness of ℓ we have $|\ell(v)| \leq \|\ell\|_{V'} \|v\|_{1,\Omega}$. Hence

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- This proves i).

Existence of Minimizer

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- $\ell \in V'$ and thus $\ell(v_j) \to \ell(v)$.
- Therefore,
$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \leq \frac{1}{2} \liminf_{j\to\infty} a(v_j, v_j) - \lim_{j\to\infty} \ell(v_j) = \liminf_{j\to\infty} \mathcal{J}(v_j).$$

$\square$

Proposition (Existence of Minimizer of $\mathcal{J}$)

The functional $\mathcal{J}$ *has at least one minimizer* $u \in K$*.*

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- Since V is Hilbert space there exists subsequence $\{v_{j_k}\}_{k \in \mathbb{N}}$ and some $u \in V$ such that $v_{j_k} \rightharpoonup u$ in V.

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- $u \in K$ since K is (weakly) closed.
- By w.l.s.c. of $\mathcal{J}$, $u \in K$ is minimizer of $\mathcal{J}$ because
$$\mathcal{J}(u) \leq \liminf_{k \to \infty} \mathcal{J}(v_{j_k}) = M = \inf_{v \in K} \mathcal{J}(v).$$

$\square$

Uniqueness of Minimizer

Proposition (Uniqueness of a Minimizer of $\mathcal{J}$)

If there exists a minimizer of the functional $\mathcal{J}$, it is unique.

Proof: $\mathcal{J}$ is strictly convex on V and thus on the convex set K. Thus, a minimizer $u \in K$ is unique.$\square$

A-Priori Stability Bound

Proposition (Stability estimate for the minimizer)

The Unique minimizer $u \in K$ of $\mathcal{J}$ satisfies the a-priori bound

$$\|u\|_{1,\Omega} \leq \frac{2\, C_K^2\, \max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}} \left(\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}\right).$$

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- Finally,

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□

Variational Inequality as an Optimality Condition

- Fréchet Derivative of $\mathcal{J}$ at $u \in V$ in direction $h \in V$ is $\mathcal{J}'(u)[h] = a(u,h) - \ell(h)$.

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- By chainrule, $\phi'(0^+) = \mathcal{J}'(u)[v-u] \geq 0$.

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- Let $u \in K$ such that $\mathcal{J}'(u)[v-u] \geq 0$ for all $v \in K$ and set $h := v - u$.

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- Let $u \in K$ such that $\mathcal{J}'(u)[v-u] \geq 0$ for all $v \in K$ and set $h := v - u$.
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- $u \in K$ minimizes $\mathcal{J}$ over K. $\square$

Problem (Signorini's Problem – Weak Formulation)

Find $u \in K$ *such that*

$$a(u, v - u) \geq \ell(v - u) \qquad \forall\, v \in K.$$

Weak Formulations / Variational Inequality

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Find $u \in K$ *such that*

$$\begin{cases} a(u,u) = \ell(u), \\ a(u,v) \geq \ell(v) \quad \forall\, v \in K. \end{cases}$$

Another Stability Estimate

Proposition (Lipschitz Dependence on the Data)

For $i \in \{1,2\}$, let $f_i \in L^2(\Omega;\mathbb{R}^d)$ and $F_i \in L^2(\Gamma_N;\mathbb{R}^d)$, and let $\ell_i \in V'$ be the corresponding load functional, i.e.

$$\ell_i(v) := \int_\Omega f_i \cdot v \,\mathrm{d}x + \int_{\Gamma_N} F_i \cdot \gamma v \,\mathrm{d}s, \qquad v \in V.$$

Moreover, let $u_i \in K$ be the unique solution of one of the weak formulations above. Then the solution depends Lipschitz continuously on the load functional, i.e.

$$\|u_1 - u_2\|_{1,\Omega} \le \frac{C_K^2}{\alpha_{\mathcal{A}}} \|\ell_1 - \ell_2\|_{V'}.$$

In particular, it depends Lipschitz continuously on the data f_i, F_i in the sense that

$$\|u_1 - u_2\|_{1,\Omega} \le \frac{C_K^2 \max\{1, C_{\mathrm{tr}}\}}{\alpha_{\mathcal{A}}} \Big(\|f_1 - f_2\|_{0,\Omega} + \|F_1 - F_2\|_{0,\Gamma_N} \Big).$$

Content

Stress Regularity

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- Therefore Green's formula is

$$\langle \sigma(u)\mathrm{n}, \gamma v\rangle_\Gamma = \int_\Omega \sigma(u) : \varepsilon(v)\,\mathrm{d}x + \int_\Omega \operatorname{div}\sigma(u)\cdot v\,\mathrm{d}x \qquad \forall\, v \in H^1(\Omega;\mathbb{R}^d).$$

A Hybrid Formulation

Problem (Signorini's Problem – Hybrid Formulation)

For given $f \in L^2(\Omega; \mathbb{R}^d)$ *and* $F \in L^2(\Gamma_N; \mathbb{R}^d)$, *find* $u \in K$ *such that* $\sigma(u) \in H(\mathrm{div}; \Omega)$ *and*

$$
\begin{cases}
-\operatorname{div} \sigma(u) = f & \text{a.e. in } \Omega, \quad (1a) \\
\langle \sigma(u)\mathrm{n}, \gamma u \rangle_\Gamma = \displaystyle\int_{\Gamma_N} F \cdot \gamma u \, \mathrm{d}s, & \quad (1b) \\
\langle \sigma(u)\mathrm{n}, \gamma v \rangle_\Gamma \geq \displaystyle\int_{\Gamma_N} F \cdot \gamma v \, \mathrm{d}s & \forall\, v \in K. \quad (1c)
\end{cases}
$$

Equivalence of Hybrid and Weak Formulation

Proposition (Equivalence of Hybrid and Weak Formulation)

Assume $\sigma(u) \in H(\mathrm{div};\Omega)$*. Then* $u \in K$ *solves the hybrid problem above if and only if it solves the weak problem*

$$\text{Find } u \in K \text{ such that} \qquad a(u, v-u) \geq \ell(v-u) \qquad \forall v \in K.$$

Key Identity for Proof

- Assume $\sigma(u) \in H(\mathrm{div};\Omega)$ and $-\operatorname{div}\sigma(u) = f$ in $L^2(\Omega;\mathbb{R}^d)$.

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$$\langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma = a(u,w) + \int_\Omega \operatorname{div}\sigma(u)\cdot w \,\mathrm{d}x = a(u,w) - \int_\Omega f\cdot w\,\mathrm{d}x.$$

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- Hence for all $w \in V$:

$$a(u,w) - \ell(w) = \langle \sigma(u)\mathrm{n}, \gamma w\rangle_\Gamma - \int_{\Gamma_N} F\cdot \gamma w\,\mathrm{d}s.$$

Proof: Hybrid $\Rightarrow$ Weak

- Subtract (1b) from (1c):

$$\langle \sigma(u)\mathrm{n}, \gamma(v-u)\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma(v-u)\,\mathrm{d}s.$$

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- Insert the key identity from the previous slide:

$$a(u, v-u) - \ell(v-u) \geq 0 \qquad \forall\, v \in K,$$

i.e. the weak formulation.

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- Thus, $-\operatorname{div} \sigma(u) = f$ in $\mathcal{D}'(\Omega; \mathbb{R}^d)$ and by $\sigma(u) \in H(\mathrm{div}; \Omega)$ we have

$$-\operatorname{div} \sigma(u) = f \qquad \text{a.e. in } \Omega.$$

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- With the key identity this is exactly (1b) since

$$0 = a(u,u) - \ell(u) = \langle \sigma(u)\mathrm{n},\, \gamma u \rangle_\Gamma - \int_{\Gamma_N} F \cdot \gamma u \,\mathrm{d}s.$$

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- Let $v \in K$. We have

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- This means

$$\langle \sigma(u)\mathrm{n}\,\gamma v\rangle_\Gamma \geq \int_{\Gamma_N} F \cdot \gamma v\,\mathrm{d}s + \underbrace{\left(\langle \sigma(u)\mathrm{n}\,\gamma u\rangle_\Gamma - \int_{\Gamma_N} F \cdot \gamma u\,\mathrm{d}s\right)}_{=0}.$$

$\square$

Normal/Tangential Splitting and Contact Trace Spaces

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- Every $w \in W_C$ admits the decomposition
$$w = w_{\mathrm{n}}\mathrm{n} + w_t \in W_{C,\mathrm{n}}\,\mathrm{n} \oplus W_{C,t}$$

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  of admissible normal traces.
- Dual cone of weakly nonpositive normal contact forces:

$$\Lambda_C := \left\{\lambda \in W'_{C,\mathrm{n}} \;\middle|\; \langle \lambda, \eta \rangle_{\Gamma_C} \geq 0 \quad \forall \eta \in W^-_{C,\mathrm{n}} \right\}.$$

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-\operatorname{div}\sigma(u) = f & \text{a.e. in } \Omega, \quad (2a)\\
\sigma_t(u) = 0 & \text{in } W'_{C,t}, \quad (2b)\\
u_{\mathrm{n}} \leq 0 & \text{a.e. on } \Gamma_C, \quad (2c)\\
\sigma_{\mathrm{n}}(u) \in \Lambda_C, & \quad (2d)\\
\langle \sigma_{\mathrm{n}}(u),\, u_{\mathrm{n}} \rangle_{\Gamma_C} = 0. & \quad (2e)
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- Assuming $|\Gamma_N| = 0$, this is equivalent to previous formulations.

Content

Further Topics